HPA – High Performance Architectures

TOP 500 SUPERCOMPUTERS REPORT

TOP 13: Isambard-AI phase 2

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***Abstract* — This project aims to describe the Isambard-AI phase 2 cluster at several levels. It takes a hierarchical approach, from the physical infrastructure down to the microarchitecture of the nodes, and validates it with benchmarks and a theoretical analysis.**

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1. Introduction

Since the invention of the first commercial microprocessor, the Intel 4004 in 1971, up to the present day, computing capacity has grown exponentially. However, this growth is now converging toward its asymptote due to the physical and thermodynamic limits that govern the semiconductor industry. Historically, during the last decades of the 20th century and the early 21st, the increase in performance rested on two theoretical pillars: Dennard scaling [1] and Moore's law [2].

Dennard scaling, formalized by Robert H. Dennard [1], established that when the physical dimensions of transistors are reduced, voltage and current must scale proportionally. This allowed the clock frequency to increase linearly with lithographic miniaturization, keeping power density per unit area constant. Nevertheless, this paradigm collapsed [3] in the early 2000s, when advances in lithography made the effects of quantum mechanics significantly apparent. Specifically, the appearance of electron tunneling prevented further reduction of the threshold voltage, unleashing unsustainable leakage currents and crushing the power wall.

In parallel, Moore's law [2], which initially predicted the doubling of transistor density every year, and later every two years [4], has suffered a severe slowdown, attributed both to the exponential increase in the cost of the equipment needed for new EUV lithographies [5] and to the approach of the atomic limits of silicon. The deceleration of this model has recently led to proposals for new scaling paradigms based on minimizing signal propagation delays by moving from 2D architectures to vertical integration [6].

Faced with the stagnation of frequency-based vertical scaling, the industry's response was the systematic exploitation of horizontal scaling and thread-level parallelism (TLP) [7]. AMD pioneered the transition to multicore x86 processors in 2005, with the introduction of the Athlon 64 X2, with a native monolithic design [8]. Two decades later, the exploitation of TLP in shared-memory systems has reached unprecedented figures, as demonstrated by the sixth generation of AMD EPYC processors, capable of integrating up to 256 cores per *socket* [9].

Despite this drastic scaling, general-purpose processors exhibit architectural inefficiencies when faced with highly parallelizable workloads. These problems call for specialized accelerators, such as graphics processing units (GPUs), which sacrifice single-thread flow control in favor of the parallel SIMT paradigm. However, contemporary requirements exceed not only the computing capacity of a single integrated circuit, but also the capacity to transfer data from memory units to compute units. The historical imbalance between the rapid growth of computing capacity and the slower growth of memory bandwidth gives rise to a systemic memory bottleneck, first presented in 1995 [10], which is becoming increasingly relevant in the training of large language models, as Gholami et al. [11] show.

To overcome this bottleneck and approach exascale computing, it is imperative to deploy heterogeneous clusters interconnected by network topologies with high bandwidth and low latency. In the geopolitical context, this massive infrastructure is critical for the training and development of large language models, in order to guarantee national technological sovereignty.

This framework of sovereign AI underpins the British government's strategic AI Research

Resource (AIRR) initiative [12]. In partnership with the Bristol Centre for Supercomputing [13], it spearheads Phase 2 of Isambard-AI, the United Kingdom's most advanced AI-centric supercomputing infrastructure

This report will analytically and systematically break down the Isambard-AI phase 2 cluster, currently ranked 13th on the TOP500 list [14].

To do so, a hierarchical architectural approach is adopted: Section 2 starts from the underlying physical infrastructure and facilities (Facility Level). Next, Section 3 addresses the structural analysis of the racks, the network interconnection topology and the storage system. Finally, Section 4 provides the microarchitectural characterization of the compute nodes, concluding with an empirical and theoretical validation through the exploration and interpretation of their results on performance benchmarks

2. Facility level system overview

At a high level, the Isambard-AI cluster consists of a total of 5448 *NVIDIA GH200* (Grace Hopper) superchips [15] distributed across 13 high-density racks, which, according to Flynn's taxonomy, makes it a MIMD system. The first rack, an HPE Cray EX2500, belongs to the Isambard-AI phase 1 supercomputer, while the other 12 racks, HPE Cray EX4000, belong to the Isambard-AI phase 2 supercomputer [16], which is broken down in this document.

The supercomputer operates at a power consumption of 5 MW, providing a maximum computing power of 216.50 PFlops in `fp64` on the Linpack benchmark [14]. Moreover, thanks to the tensor architecture provided by the NVIDIA GPUs, it is theoretically capable of reaching 21 EFlops/s in `fp8` format [16].

2.1. Brief history of the project

The conception of the system responds directly to data sovereignty and national security policies. Unlike pan-European infrastructures developed under the EuroHPC consortium, the British government developed its own strategy under the *AI Research Resource* (AIRR) program [12]. This government initiative involved an initial injection of 225 million pounds in autumn of 2023 [17].

Execution of the project was handed off to the University of Bristol, due to its proven experience with the GW4 project [18]. Specifically, the cluster is overseen by Prof. Simon McIntosh-Smith, director of the *Bristol Centre for Supercomputing* (BriCS), and Dr. Sadaf Alam, Chief Technology Officer (CTO) of the same institution. The system was assembled collaboratively by Hewlett Packard Enterprise (HPE) and NVIDIA, and despite the logistical complexity of the project, phase 1 was completed in record time, going from contract signing to operational status in less than 4 months.



Figure 1. Aerial view of the center during construction of the data center

2.2. Facilities

The physical system is housed in the *National Composites Centre* (NCC), located in the *Bristol and Bath Science Park*. It shares the site with phase 1 of Isambard-AI, as well as with the Isambard 3 cluster, based exclusively on NVIDIA Grace CPUs. Due to strict space limitations, with only 550 square meters available ($15.6 \times 35m$) [16], the system could not be deployed in a conventional clean room. Instead, a self-contained solution was implemented that encloses the nodes, electrical transformation, and pumping stations within the enclosure itself. To achieve this, a modular data center architecture was used, which made it possible to reduce by 72% the emissions derived from its construction compared with conventional data centers [16].

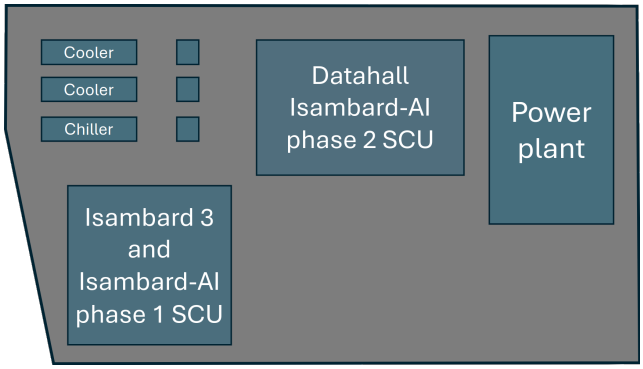


Figure 2. Diagram of the system layout [16]

To dissipate the high density of thermal energy generated, which exceeds 400 kW per rack, conventional air cooling is not feasible. Instead, Direct Liquid Cooling is used, a solution standardized in the HPE Cray EX chassis [19]. This approach not only prevents overheating of the components, but its efficiency also makes the system a referent of the Green AI paradigm: its twin supercomputer, Isambard AI-phase 1, ranks 4th in the June 2026 Green AI ranking [20]. This sustainability milestone is especially notable for having been achieved with accelerators from the Hopper generation, which exhibit higher nominal energy consumption per flop than more lithographically advanced architectures, such as Blackwell or Vera Rubin. The system has a nominal Power Usage Effectiveness (PUE) of 1.08, meaning that only 8% of the energy consumed goes to peripheral cooling, markedly better than the figure common for air-cooled systems, which hovers around 1.54–1.56 [21].

The environmental impact of the machine's consumption is minimized through its electricity procurement policy. Although its high consumption of 5 MW required a high-capacity direct connection to the British national grid, 100% of the energy supplied to the system comes from renewable sources in the UK, covering not only wind, solar and hydroelectric systems but also battery storage systems [22].

3. Rack, Interconnection & Storage

The rack-level and interconnection architecture is designed hierarchically to maximize compute density and minimize latency, responding to the operational demands of contemporary artificial intelligence workloads.

At the rack level, each of the 12 HPE Cray EX4000 [16] cabinets available contains a total of 110 compute nodes, which provides a total of 440 NVIDIA GH200 superchips per cabinet.

Within each rack, the basic unit is the HPE Cray EX254n *blade* accelerator. It houses two independent compute nodes, each of which has 4 NVIDIA Grace processors of 72 cores (ARM Neoverse V2 architecture), as well as 4 NVIDIA H100 GPUs (Hopper architecture). Consequently, each node has 288 ARM v9 cores and 4 H100 GPUs.

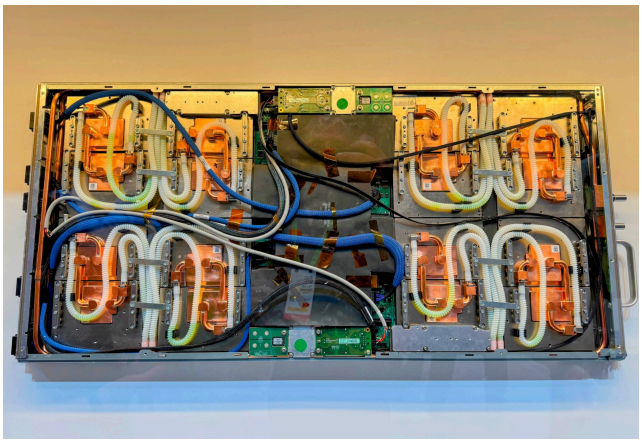


Figure 3. HPE Cray EX254n blade chassis

At the cluster interconnect level, network orchestration is delegated to the HPE Slingshot 11 architecture [23] The HPE Cray EX system uses an orthogonal cableless coupling [19]: compute blades are inserted from the front of the chassis, while switch blades are inserted from the rear, intersecting at a midplane. This design fundamentally serves 3 goals:

- **Improve signal integrity:** allowing greater bandwidth.
- **Minimize interconnect latencies:** the short trace layout minimizes propagation delays.
- **Minimize energy expenditure:** reducing the amount of energy needed per transmitted byte.

Each node uses 4 network cards, offering a total bandwidth of 800 Gb/s of simultaneous input and output [13] per node toward the main network. For inter-node connections, 64-port switches are used, offering a total of 25.6 Tb/s.

At the macroscopic level, the network topology used is of the *Dragonfly* type [24]. This design is based on all-to-all meshes of small groups of nodes, which are subsequently interconnected with each other; this makes it possible to guarantee a diameter of 3 hops [23] for ≤ 37120 endpoints.

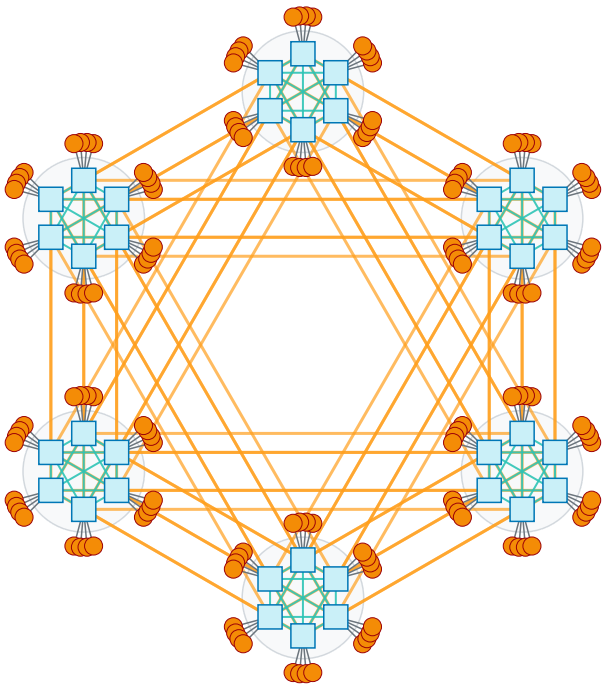


Figure 4. Example of a dragonfly topology

Although the exact implementation of this topology in the Isambard-AI phase 2 cluster is unknown, it is possible to infer it using other systems as a reference, such as the *Alps* system, belonging to the *Swiss National Supercomputing Centre* [25]. A possible implementation of this architecture for each switch is shown in Figure 5.

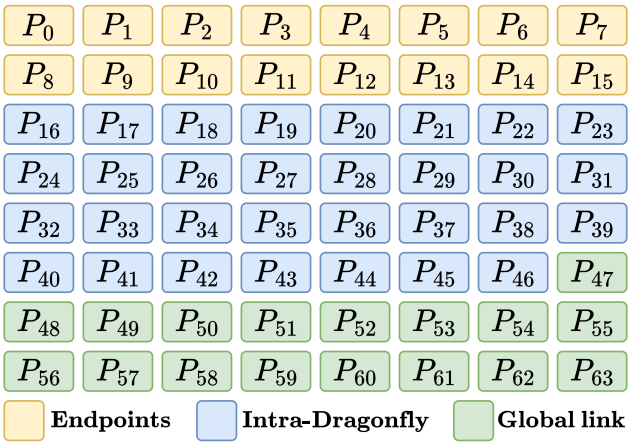


Figure 5. Distribution of the switches in the *Alps* supercomputer

3.1. Storage

Storage responds to requirements that differ from those of classical HPC simulation. Training requires high-speed reads of datasets and periodic writes of checkpoints, while inference demands a large number of small, parallel reads. For this reason, a traditional parallel file system alone is not enough [16]. To cover this, Isambard-AI phase 2 combines three tiers.

The first is an all-flash HPE ClusterStor E1000 storage with Lustre, made up of 1056 NVMe SSDs of 30.72 TB each, with about 20.3 PiB usable. The expected aggregate performance reaches up to 2500 GB/s for reads and 1980 GB/s for writes, with 35 million random 4 KB read IOPS. The second is a multiprotocol, multi-tenant SDS platform based on VAST Data, with about 3.56 PB usable, designed for AI workloads and for isolating sensitive data between projects. The third is local 3.84 TB SSDs on a subset of nodes, intended for small or sensitive data.

4. Node microarchitecture

Each node of the Isambard-AI phase 2 cluster brings together multiple NVIDIA GH200 superchips, each of which integrates its own Grace processor and H100 GPU, communicating via NVLink-C2C (Chip to Chip). The main characteristics of these components are described throughout this section.

4.1. Node composition

At the macro level, each node has 4 GH200 superchips, which add up to 288 ARM cores, 67584 CUDA cores, 512 GB of LPDDR5X memory and 384 GB of HBM3 memory, distributed equally among the superchips [13]. The tight integration allows each CPU or GPU to access any RAM or VRAM address of any other chip [15].

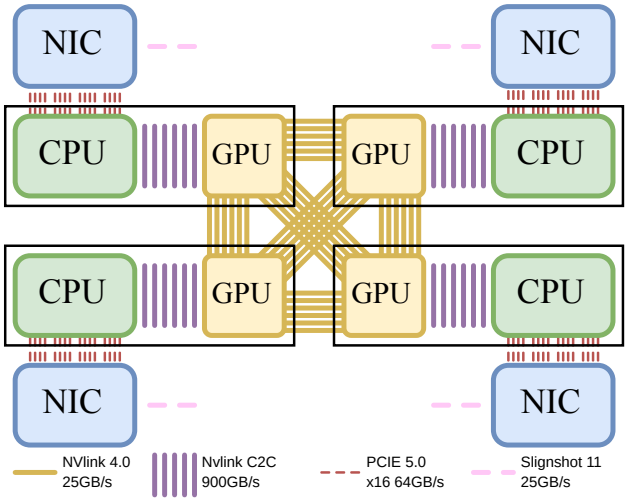


Figure 6. Diagram of an Isambard-AI phase 2 node: four GH200 superchips, each with its Grace CPU, H100 GPU, LPDDR5X and HBM3 memories, its Slingshot-11 network card and the intra-superchip NVLink-C2C links. Reference diagram based on [25].

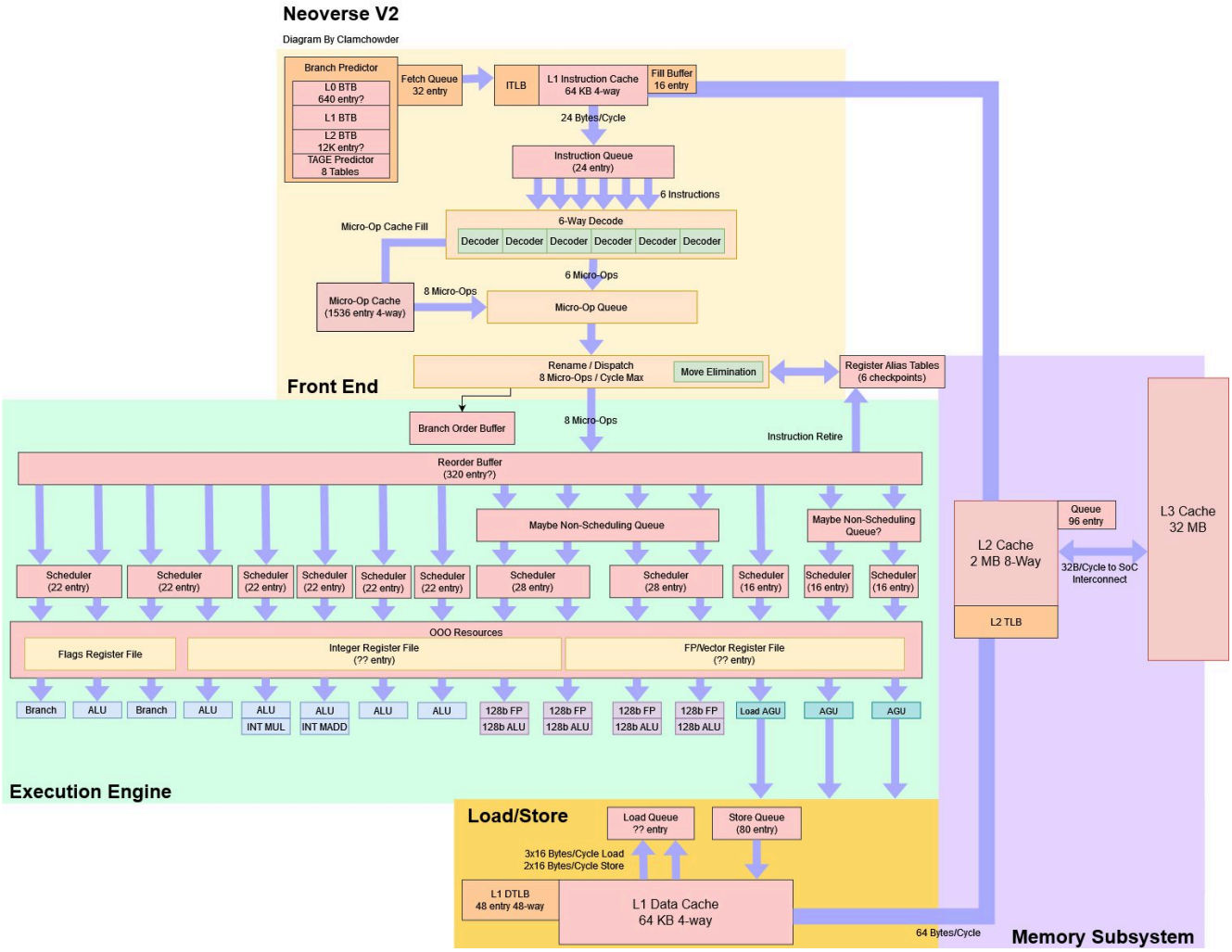


Figure 7. Microarchitectural pipeline of a processor based on the Neoverse V2 specification [27]. It does not strictly correspond to the processor described in this section.

At the intra-node interconnect level, the GPUs are connected in an all-to-all mesh based on NVLink 4.0 technology, with 6 links per connection, offering a bandwidth of 150 GB/s per direction [25]. For its part, CPU-GPU communication within the same superchip is carried out via NVLink-C2C, offering a total of 450 GB/s per direction [15], far surpassing the 64 GB/s per direction offered by PCIe 5.0. This reduces the minimum arithmetic intensity required for GPU execution to be worthwhile compared with the CPU.

Regarding latency, according to the analysis by Fusco et al. [26], this unified-memory NUMA architecture shows asymmetric behavior depending on where the data resides.

The Grace CPU accesses its local LPDDR5X memory with a latency of 170.0 ns, while accessing the HBM3 of its paired GPU through C2C raises the cost to 798.8 ns. The Hopper GPU accesses its local HBM3 in 344.2 ns and needs 817.8 ns to access the CPU's LPDDR5X through the C2C link.

Peer-to-peer communications between superchips reflect routing overhead: a GPU accessing the HBM of another GPU incurs a latency of 1022.9 ns, while a CPU accessing the LPDDR5X of another CPU waits 520 ns.

4.2. The CPU

The Grace processor is a 72-core CPU with frequencies of up to 3.1 GHz, based on the ARM Neoverse V2 microarchitecture (ARMv9-A), which was presented at the Hot Chips 2023 conference [27]. It is a superscalar core, with great capacity to exploit instruction-level parallelism (ILP) thanks to out-of-order execution and its width of 8 μ ops per cycle. However, it does not implement simultaneous multithreading (SMT), so each core executes a single physical thread. According to Flynn's taxonomy, it is a MIMD system whose cores can execute SIMD and SISD instructions.

As seen in Figure 7, the processor's frontend can decode up to 6 instructions per cycle, as well as dispatch 8, relying on a μop cache of 1536 entries,

4-way set associative. It also has a TAGE branch predictor [28].

The backend has 6 scalar integer units (ALU), two of which perform `mul` operations and one of which also performs `fma`. It also has 2 branch units, 4 128-bit FP/SIMD ports, and 3 address generation units (AGU). Two of the AGUs handle loads and stores, while the third handles only loads, which theoretically allows up to 3 memory operations of 16 bytes per cycle.

Memory in this processor follows the hierarchy shown below:

Type	Size	Associativity	Bytes/cycle	Bandwidth
Registers	Not published	—	—	—
L1d	64 KB/core	4-way	48	144 GB/s/core
L2	1 MB/core	8-way	64	192 GB/s/core
L3	114 MB	—	—	>2 TB/s [29]
LPDDR5X	128 GB	—	—	512 GB/s

Table 1. Memory hierarchy of the Grace (CPU) processor of the GH200 superchip.

Communication between threads, exploited through thread-level parallelism (TLP), is carried out through a coherency mesh based on a 6×7 grid, shown in Figure 8. With Manhattan distance, the maximum number of hops between nodes is $6 + 7 = 13$.

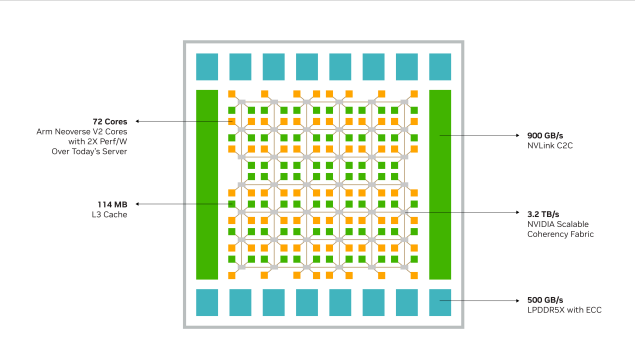


Figure 8. Block diagram of the Grace (CPU) processor of the GH200: 72 Neoverse V2 cores and 114 MB of distributed L3 cache over the coherency mesh [30].

The 72-core processor operates at 3.0 GHz using 4 128-bit SIMD pipelines per core, belonging to the ARMv9-A family (SVE2/NEON). Its computational capabilities are broken down below.

- **`fp64`:** Each pipeline includes 2 lanes performing one `FMA` (2 operations). This yields 16 FLOP/cycle per core. The total peak is $72 \times 16 \times 3.0$ GHz $= 3.456$ TFLOPS.
- **`fp32`:** Each pipeline includes 4 lanes performing one `FMA` (2 operations). This yields 32 FLOP/cycle per core. The total peak is $72 \times 32 \times 3.0$ GHz $= 6.912$ TFLOPS.
- **`bf16`:** The `BFDOT` instruction accumulates 2 products per 32-bit lane, equivalent to 8 `mac` (16 FLOP) per instruction. This results in 64 FLOP/cycle per core. The total peak is $72 \times 64 \times 3.0$ GHz ≈ 13.8 TFLOPS.
- **`int8`:** The `SDOT` instruction accumulates 4 products per 32-bit lane, equivalent to 16 `mac` (32 operations) per instruction. This results in 128 OPS/cycle per core. The total peak is $72 \times 128 \times 3.0$ GHz ≈ 27.6 TOPS.

4.3. The GPU

Each superchip incorporates an H100 GPU with 132 enabled streaming multiprocessors (SM), out of a total of 144 on the die [29]. According to Flynn's taxonomy, the GPU, since it has several SMs, generally behaves as a MIMD model, while within each SM it is a degenerate case of SIMD, namely SIMT. Bewlo, a description in two parts is provided: first the memory hierarchy, then its computational capabilities.

4.3.1. Memory hierachy

Type	Size	Bytes/cycle	Bandwidth	Latency
Registers	256 KB/SM	—	—	1 cycle
L1	256 KB/SM	128/SM	≈ 33 TB/s	≈ 23–41 cycles
L2	2×25 MB	—	≈ 5.5 TB/s	≈ 200–263 cycles
HBM3	96 GB	—	≈ 4 TB/s	≈ 290–480 cycles

Table 2. Memory hierarchy of the H100 (Hopper) GPU of the GH200 superchip. Latency data from [31]

The SM registers occupy 256 KB, shared among the resident warps: each thread can use up to 255 32-bit registers (about 1 KB), which amounts to up to 32 KB per warp of 32 threads [32].

The L1 has 256 KB per SM, offering a bandwidth of 128 B/cycle per SM [33], which allows it to reach about 33 TB/s aggregated over the 132 SMs.

The 50 MB L2 cache is shared by all SMs and is divided into two partitions of 25 MB each. Accesses to the far partition cost roughly twice as much as accesses to the near one. Chips and Cheese measured [29] more than 5.5 TB/s of read bandwidth on the near partition, but on an H100 PCIe with 114 SMs, so for the SXM variant this is an estimate.

La HBM3 del GH200 ofrece 96 GB y 4 TB/s, a través de un bus de 6144 bits.

Finally, Hopper's thread block clusters allow co-scheduling up to 16 blocks on adjacent SMs. A block can access the shared memory of another through an SM-to-SM network [32], which accord-

ing to NVIDIA reduces exchange latency by about 7 times.

4.3.2. Compute per SM

Each SM contains 128 `fp32` cores, 64 `fp64` cores and 4 fourth-generation tensor cores. Figure 9 shows the organization of an SM of the H100 GPU.

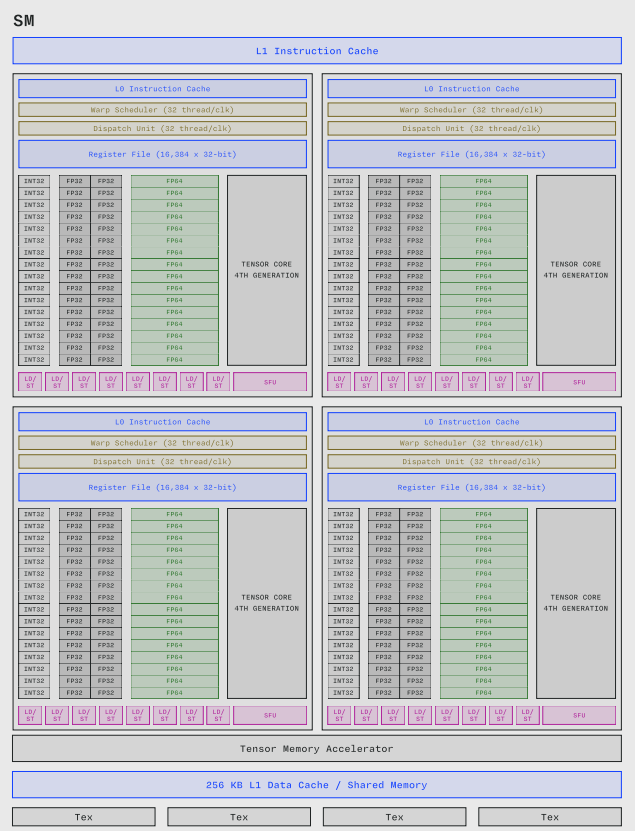


Figure 9. Diagram of the SM of the Hopper (H100) architecture. Taken from [34]

The compute of each SM depends on the data type. In the CUDA cores, each `fp32` core executes one `fma` per cycle, which gives 128 `fma`, or 256 FLOP, per SM and cycle. In `fp64`, the CUDA cores perform 64 FMA per cycle, that is, 128 FLOP.

The tensor cores are the ones that stand out most in lower-precision formats. Table 3 summarizes these values together with the arithmetic intensity, which is studied in the final table.

Type	Unit	FLOP/c/SM	TFLOPS	AI/L1	AI/HBM3
`fp64`	CUDA	128	≈ 34	1	≈ 8.5
`fp32`	CUDA	256	≈ 67	1	≈ 16.75
`fp64`	Tensor	256	≈ 67	2	≈ 16.75
`tf32`	Tensor	2048	≈ 494	16	≈ 123.5
`FP16/BF16`	Tensor	4096	≈ 990	32	≈ 247.5
`FP8/INT8`	Tensor	8192	≈ 1979	64	≈ 494.75

Table 3. Operations per cycle and per SM, peak of the 132-SM system, and minimum arithmetic intensity for compute-bound workloads.

The intensities explain where the bottleneck lies. A CUDA `fp64` kernel already needs about 8.5 FLOP per byte of HBM3 to saturate the compute, while tensor FP16 needs about 247. Language model training workloads reuse each value heavily and can reach that intensity, but low-intensity kernels remain limited by memory.

4.4. Benchmarking

The results are divided into two blocks: the Grace CPU, measured by Phoronix [35], and the H100 GPU, for which there is only manufacturer data and third-party data with different configurations.

4.4.1. CPU Grace

Phoronix compares the 72 cores of the GH200 with the AMD EPYC 9654 and the Intel Xeon 8592+, under Ubuntu 23.10, Linux 6.5 and GCC 13. The tested system is a GPTshop.ai desktop with 480 GB of LPDDR5X, not the Isambard-AI cluster: since they share the CPU, the results characterize the processor and not the complete system.

Benchmark	Workload	Grace (GH200)	EPYC 9654	Xeon 8592+
HPCG (GFLOP/s)	memory-bound	32.13	43.97	35.42
Graph500 (TEPS)	irregular access	1.24×10^9	0.816×10^9	1.199×10^9
LULESH z/s	hydrodynamics	23185	23701	39468

Table 4. Benchmark results of the Grace CPU against reference processors [35]

In HPCG, Grace obtains 27% less than the EPYC 9654, despite its peak bandwidth of 512 GB/s. In Graph500, with irregular accesses, it outperforms both x86 processors. In LULESH it is on par with the EPYC, while the Xeon archiecves almost 1.7x that result.

4.4.2. GPU H100

For the GPU, NVIDIA publishes a comparison of the 144 GB GH200 variant against an x86 reference and an x86 system with H100 [36]. The chart does not detail the reference configuration, so these are manufacturer figures.

Workload	Type	x86 + H100	GH200 (144 GB)
ICON	HPC	6×	8×
MILC	HPC	27×	41×
Quantum Fourier Transform	HPC	61×	94×
DLRM inference (100 GB)	AI	9×	18×
IVF-PQ vector search	AI	25×	33×
GNN training (500 GB)	AI	5×	40×

Table 5. Speedup relative to an x86 reference (1.0×), according to NVIDIA [36]

As an independent contrast, Databricks measured on nodes of 8 H100 SXM 80 GB against A100 80 GB [37]. BF16 training is 2.2× faster, and with FP8 it reaches up to 3.3×. Matrix multiplications reach close to 5× against a theoretical maximum of 6.3×, while memory-bound operations improve by at most 1.6×, in line with the bandwidth advantage of the H100 over the A100.

These figures correspond to the GPU in another configuration and from another software generation, so they serve as a reference for the GPU's behavior, not as a measurement of the GH200 in the cluster.

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